

Deriving 4H-Silicon Carbide's Spin Hamiltonian

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The Spin Hamiltonian is a combination of the Zeeman effect, Hyperfine Interactions, the Zero-Field Splitting Effect, and the Exchange Interaction.

$$\hat{H}_0 = \hat{H}_Z + \hat{H}_{HF} + \hat{H}_{ZFS} + \hat{H}_{EX}$$

Because Silicon Carbide (SiC) has no nuclei with nuclear spin quantum number $I > \frac{1}{2}$, there is no Nuclear Quadrupole Interaction Hamiltonian.

The goal of this paper is to provide a thorough overview and derivation of each Hamiltonian term. Finally, a complete Hamiltonian description will be provided along with all unknown constants that need to be derived from experimental data.

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1 The Zeeman Effect

A spinning charged particle is a magnetic dipole. Its magnetic dipole moment, $\vec{\mu}$, is proportional to its spin angular momentum, $\vec{S}$:

$$\vec{\mu} = \gamma \vec{S};$$

the proportionality constant, γ , is the gyromagnetic ratio. From the Dirac equation it can be shown

$$\vec{\mu} = \gamma \vec{S} = -g \frac{q}{2m_e} \vec{S} = -\frac{g\mu_B}{\hbar} \vec{S} \quad (1)$$

where $g \approx 2.0023$ is the Lande g factor and $\mu_B \approx 9.2 \cdot 10^{-24} J/T$ is the Bohr Magnetron.

When a magnetic dipole is placed in a magnetic field $\vec{B}$, it experiences a torque, $\vec{\mu} \times \vec{B}$, which tends to line it up parallel to the field like a compass. The energy associated with this torque is

$$H = -\vec{\mu} \cdot \vec{B} = -\gamma \vec{B} \cdot \vec{S} \quad (2)$$

Therefore, the Zeeman Hamiltonian,

$$\hat{H}_Z = -\mu_0 \cdot \vec{B}_0 = \frac{g\mu_B}{\hbar} \vec{S} \cdot \vec{B}_0 = \frac{g\mu_B}{\hbar} B_0 S_z$$

where $S_z = \frac{\hbar}{2} \hat{\sigma}_z$ and σ_z is the Pauli-z spin matrix. Applying $\hat{H}_Z$ on the state $|s, m_s\rangle$, we find

$$\begin{aligned} \hat{H}_Z |s, m_s\rangle &= \frac{g\mu_B}{\hbar} B_0 \cdot S_z |s, m_s\rangle \\ &= \frac{g\mu_B}{\hbar} B_0 \cdot m_s \hbar |s, m_s\rangle \\ &= m_s g \mu_B B_0 |s, m_s\rangle. \end{aligned}$$

To generalize to an anisotropic g , we replace with a 3×3 g tensor,

$$\hat{H}_Z = \mu_B \vec{S} \cdot g \cdot \vec{B}_0$$

g is symmetric and diagonalizable via the defect's *principle axis*.

$$g_{\text{diag}} = \begin{pmatrix} g_x & 0 & 0 \\ 0 & g_y & 0 \\ 0 & 0 & g_z \end{pmatrix}$$

However, for fitting purposes, setting $g \approx g_{\text{free } e^-} = 2.0023$ is acceptable. Hence this Hamiltonian is simple:

$$\hat{H}_Z |s, m_s\rangle = m_s g \mu_B B_0 |s, m_s\rangle.$$

2 Hyperfine Interaction

2.1 Vector Potential $\vec{A}$

From Maxwell's equations, we have

$$\nabla \cdot \vec{B} = 0, \quad \nabla \times B = \mu_0 \vec{J}.$$

Because $\nabla \cdot B = 0$, we set $\vec{B} = \nabla \times \vec{A}$. We impose the Coulomb gauge $\nabla \cdot \vec{A} = 0$. Then

$$\nabla^2 \vec{A} = -\mu_0 \vec{J} \quad (3)$$

2.2 Current density $\vec{J}$

We model the localized nuclear magnetic moment $\vec{\mu}_n$ at the origin by an effective current density

$$\vec{J}_n(\vec{r}') = \nabla' \times [\vec{\mu}_n \delta^3(\vec{r}')].$$

The general solution for A (Equation 3) is

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int d^3r' \frac{\vec{J}_n(\vec{r}')}{|\vec{r} - \vec{r}'|} = \frac{\mu_0}{4\pi} \vec{\mu}_n \times \nabla \left(\frac{1}{r} \right) = \frac{\mu_0}{4\pi} \frac{\vec{\mu}_n \times \vec{r}}{r^3}.$$

Therefore,

$$\vec{B}(r) = \nabla \times \vec{A} = \frac{\mu_0}{4\pi} \nabla \times \left(\frac{\vec{\mu}_n \times \vec{r}}{r^3} \right). \quad (4)$$

Using the identity

$$\nabla \times (\vec{a} \times \vec{r} f(r)) = \vec{a} \nabla \cdot (\vec{r} f) - (\vec{a} \cdot \nabla)(\vec{r} f),$$

with $\vec{a} = \vec{\mu}_n$ and $f(r) = 1/r^3$, we get the canonical expression for the magnetic field from a dipole from classical electrodynamics:

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi r^3} [3(\vec{\mu}_n \cdot \hat{r})\hat{r} - \vec{\mu}_n] + \frac{2\mu_0}{3} \vec{\mu}_n \delta^3(\vec{r}). \quad (5)$$

2.3 Hamiltonian Density

A magnetic moment $\vec{\mu}_e$ at $\vec{r}$ has energy (Equation 2)

$$U(\vec{r}) = -\vec{\mu}_e \cdot \vec{B}(\vec{r}).$$

Therefore, from Equation 5 we separate both terms,

$$U_{\text{dipolar}}(\vec{r}) = -\vec{\mu}_e \cdot \left[\frac{\mu_0}{4\pi r^3} (3(\vec{\mu}_n \cdot \hat{r})\hat{r} - \vec{\mu}_n) \right],$$

$$U_{\text{contact}}(\vec{r}) = -\vec{\mu}_e \cdot \left[\frac{2\mu_0}{3} \vec{\mu}_n \delta^3(\vec{r}) \right].$$

We promote the magnetic moments to their quantum operators:

$$\vec{\mu}_e = -g_e \mu_B \frac{\hat{S}}{\hbar}, \quad \mu_B = \frac{e\hbar}{2m_e},$$

$$\vec{\mu}_n = g_n \mu_N \frac{\hat{I}}{\hbar}, \quad \mu_N = \frac{e\hbar}{2m_p}.$$

Therefore, the Hamiltonian of the electron, in the magnetic field due to the nuclei's magnetic dipole moment at $\vec{r}$ (Equation 5), is $H = -\vec{\mu} \cdot \vec{B}$,

$$\hat{H}(\hat{r}) = \frac{\mu_0}{4\pi} \frac{g_e \mu_B g_n \mu_N}{\hbar^2} \left(\frac{\vec{S} \cdot \vec{I} - 3(\vec{S} \cdot \hat{r})(\vec{I} \cdot \hat{r})}{r^3} + \frac{8\pi}{3} \delta^3(\vec{r}) \vec{S} \cdot \vec{I} \right). \quad (6)$$

If the electron occupies spatial wave function $\psi(\vec{r})$, the full hyperfine Hamiltonian is

$$\hat{H}_{HF} = \int d^3r |\psi(\vec{r})|^2 \vec{H}(\vec{r}).$$

To solve this, we have

$$\hat{H}(\vec{r}) = C \left[\frac{8\pi}{3} \delta^3(\vec{r}) (\vec{S} \cdot \vec{I}) + \frac{1}{r^3} (\vec{S} \cdot \vec{I} - 3(\vec{S} \cdot \hat{r})(\vec{I} \cdot \hat{r})) \right],$$

where

$$C = \frac{\mu_0}{4\pi} \frac{g_e \mu_B g_n \mu_N}{\hbar^2}, \quad \hat{r}_i = \frac{r_i}{r}.$$

Writing the dipolar piece in index form,

$$\hat{H}(\vec{r}) = C \left[\frac{8\pi}{3} \delta^3(\vec{r}) S_k I_k + \frac{1}{r^3} (\delta_{k\ell} - 3\hat{r}_k \hat{r}_\ell) S_k I_\ell \right], \quad (7)$$

where we sum over repeated indices. Now to integrate, we have

$$\hat{H}_{HF} = \int d^3r |\psi(\vec{r})|^2 \vec{H}(\vec{r}) = \int d^3r \rho(\vec{r}) \hat{H}(\vec{r}).$$

where $\rho(r) = |\psi(\vec{r})|^2$ is the spatial probability density. Splitting Equation 7 into contact and dipolar integrals:

$$\hat{H}_{HF} = C \left[\frac{8\pi}{3} S_k I_k \int d^3 r \rho(\vec{r}) \delta^3(\vec{r}) + S_k I_\ell \int d^3 r \rho(\vec{r}) \frac{\delta_{k\ell} - 3\hat{r}_k \hat{r}_\ell}{r^3} \right].$$

The contact term (isotropic) reduces to

$$\begin{aligned} \frac{8\pi}{3} S_k I_k \int d^3 r \rho(\vec{r}) \delta^3(r) &= \frac{8\pi}{3} S_k I_k \rho(0) \\ &= \frac{8\pi}{3} |\psi(0)|^2 S_k I_k \\ &= A_{\text{iso}} \delta_{k\ell} S_k I_\ell, \quad A_{\text{iso}} = C \frac{8\pi}{3} |\psi(0)|^2. \end{aligned}$$

To solve the dipolar term, we define

$$A_{k\ell}^{\text{dip}} = C \int d^3 r \rho(\vec{r}) \frac{\delta_{k\ell} - 3\hat{r}_k \hat{r}_\ell}{r^3}. \quad (8)$$

Then the term reduces to $S_k I_\ell A_{k\ell}^{\text{dip}}$. We define the A_{ij} tensor as

$$A_{ij} = A_{\text{iso}} \delta_{ij} + A_{ij}^{\text{dip}}.$$

Therefore

$$\hat{H}_{HF} = \hat{S} \cdot A \cdot \hat{I} = \hat{S}_i A_{ij} \hat{I}_j.$$

For multiple nuclei, like in SiC, we sum over every j inequivalent nucleus.

$$\hat{H}_{HF} = \sum_j \hat{\vec{S}} \cdot A_j \cdot \hat{\vec{I}}_j.$$

The dipolar integral in A_{ij}^{dip} is often evaluated by choosing the defect's principal axis frame to diagonalize.

From Equation 18, the Cartesian spin operators are shown to be the sum or difference of the raising or lowering operators. This relation also holds true for Cartesian nuclear spin operators acting on nuclear spin. Therefore, with two nuclei, like in SiC, the Hyperfine Hamiltonian acting on a state $|s, m\rangle |m_{I_1}\rangle |m_{I_2}\rangle$ can be written as

3 Zero-Field Splitting

The Zero-Field Splitting Hamiltonian is the dipole-dipole interaction between a paramagnetic defect electron and the carrier electron. This is incredibly similar to the Hyperfine interaction between a nearby nucleus and the carrier electron. We can start from Equation 5

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi r^3} [3(\vec{\mu}_n \cdot \hat{r})\hat{r} - \vec{\mu}_n] + \left(\frac{2\mu_0}{3} \vec{\mu}_n \delta^3(\vec{r}) \right).$$

However, here we remove the second term in the large parenthesis. This was the contact (δ -function) term only for a pointlike nucleus – in Hyperfine, the nucleus sits at a point, so when we take the curl of $\vec{A}$, we pick up the singular term $\frac{2\mu_0}{3} \vec{\mu}_n \delta^3(\vec{r})$.

However, for electron-electron interactions, two electrons never coincide exactly, so there is no δ -contact term. Only the $1/r^3$ dipolar piece survives from taking the curl. Hence, we have

$$\vec{B}_1(\vec{r}) = \frac{\mu_0}{4\pi r^3} [3(\vec{\mu}_1 \cdot \hat{r})\hat{r} - \vec{\mu}_1], \quad \hat{r} = \frac{\vec{r}}{r}, \quad (9)$$

where $\vec{\mu}_1$ is the magnetic moment of electron 1 and $\vec{B}_1(\vec{r})$ is the magnetic field of electron 1 at point $\vec{r}$. A second dipole $\vec{\mu}_2$ located at $\vec{r}$ has energy in the field of dipole 1:

$$U = -\vec{\mu}_2 \cdot \vec{B}_1(\vec{r}) = \frac{\mu_0}{4\pi r^3} [-\mu_1 \cdot \mu_2 + 3(\mu_1 \cdot \hat{r})(\mu_2 \cdot \hat{r})].$$

Now we can promote $U \rightarrow \hat{H}_{ZFS}$ by using the electron magnetic moment operator (Equation 1):

$$\hat{\mu}_i = -g_e \mu_B \frac{\hat{S}_i}{\hbar}, \quad \mu_B = \frac{e\hbar}{2m_e}, \quad g_e \approx 2.0023.$$

Therefore, replacing $\vec{\mu}_{1,2}$ by $\hat{\mu}_{1,2}$, the prefactor becomes

$$-\frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2}.$$

Therefore, we have

$$\begin{aligned} \hat{H}_{ZFS} &= -\frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \frac{1}{r^3} \left[-\hat{S}_1 \cdot \hat{S}_2 + 3(\hat{S}_1 \cdot \hat{r})(\hat{S}_2 \cdot \hat{r}) \right] \\ &= \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \frac{1}{r^3} \left[\hat{S}_1 \cdot \hat{S}_2 - 3(\hat{S}_1 \cdot \hat{r})(\hat{S}_2 \cdot \hat{r}) \right]. \end{aligned}$$

We can simplify this into an analogous form of the Hyperfine Hamiltonian. We have the equalities

$$\begin{aligned} (\vec{S}_1 \cdot \vec{S}_2) &= \sum_{i \in \{x,y,z\}} S_{1i} S_{2i} \\ (\vec{S}_1 \cdot \hat{r})(\vec{S}_2 \cdot \hat{r}) &= \left(\sum_i S_{1i} \hat{r}_i \right) \left(\sum_j S_{2j} \hat{r}_j \right) \\ &= \sum_{i,j} S_{1i} (\hat{r}_i \hat{r}_j) S_{2j} \\ &= \sum_{i,j} S_{1i} \frac{r_i r_j}{r^2} S_{2i}. \end{aligned}$$

Hence,

$$\begin{aligned}
\hat{H}_{ZFS} &= \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \frac{1}{r^3} \left(\sum_i S_{1i} S_{2i} - 3 \sum_{i,j} S_{1i} \frac{r_i r_j}{r^2} S_{2j} \right) \\
&= \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \left(\sum_i \frac{\delta_{i,j}}{r^3} S_{1i} S_{2j} - 3 \sum_{i,j} \frac{r_i r_j}{r^5} S_{1i} S_{2j} \right) \\
&= \sum_{i,j} S_{1i} \left[\frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \left(\frac{\delta_{i,j}}{r^3} - \frac{3r_i r_j}{r^5} \right) \right] S_{2j}.
\end{aligned}$$

We define the tensor D_{ij}

$$D_{ij}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right).$$

Then we get a similar equation to Hyperfine Hamiltonian:

$$\hat{H}_{ZFS} = \sum_{i,j \in \{x,y,z\}} S_{1i} D_{ij}(\vec{r}) S_{2i} = \hat{\vec{S}} \cdot \hat{\vec{D}} \cdot \hat{\vec{S}}. \quad (10)$$

We can also diagonalize $\hat{\vec{D}}$ via the principle axis. Additionally, defining $K_{ij}(\vec{r})$,

$$D_{ij}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} \left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5} \right) = \frac{\mu_0}{4\pi} \frac{(g_e \mu_B)^2}{\hbar^2} K_{ij}(\vec{r}),$$

and taking the trace,

$$\sum_i K_{ii} = \sum_i \frac{1}{r^3} - 3 \sum_i \frac{r_i^2}{r^5} = \frac{3}{r^3} - \frac{3}{r^2} (r_x^2 + r_y^2 + r_z^2) = 0.$$

Therefore, D_{ij} is traceless. So if we diagonalize via the principle axis, we have

$$D_{ij} = \begin{pmatrix} D_x & 0 & 0 \\ 0 & D_y & 0 \\ 0 & 0 & D_z \end{pmatrix}, \quad D_x + D_y + D_z = 0.$$

And because the Silicon Vacancy $V_{\text{Si}}^- \in C_{3V}$ point group, we have $D_x = D_y$. We can parameterize to fewer unknown variables.

$$\begin{cases} 2D_x + D_z = 0 \\ D_x = D_y \end{cases} \Rightarrow D_x = D_y = -\frac{1}{2}D_z.$$

Define D, E such that

$$D = \frac{3}{2}D_z \quad E = \frac{1}{2}(D_x - D_y) = 0.$$

Then, we have

$$D_{\text{diag}} = \begin{pmatrix} D_x & 0 & 0 \\ 0 & D_y & 0 \\ 0 & 0 & D_z \end{pmatrix} = \begin{pmatrix} -\frac{D}{3} + E & 0 & 0 \\ 0 & -\frac{D}{3} - E & 0 \\ 0 & 0 & \frac{2}{3}D \end{pmatrix}.$$

However, more accurately, we integrate. D is analogous to A in Equation 8 and describes the dipole-dipole interaction between two electron spins instead of electron-nuclear. We have

$$D_{ij} = \frac{\mu_0}{4\pi} g_a g_b \mu_B^2 \left\langle \psi_0 \left| \frac{3r_i r_j - \delta_{ij} r^2}{r^5} \right| \psi_0 \right\rangle,$$

which is the exact same as Equation 8. Both Hamiltonians are derived analogously. Thus,

$$\hat{H}_{ZFS} = \hat{\vec{S}} \cdot \vec{D} \cdot \hat{\vec{S}}.$$

We can expand,

$$\hat{H}_{ZFS} = \hat{\vec{S}}_a \cdot \vec{D} \cdot \hat{\vec{S}}_b \quad (11)$$

$$= \left(-\frac{D}{3} + E \right) \hat{S}_x^2 - \left(\frac{D}{3} + E \right) \hat{S}_y^2 + \frac{2}{3} D \hat{S}_z^2 \quad (12)$$

$$= D \left(\hat{S}_z^2 - \frac{1}{3} (\hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2) \right) + E(\hat{S}_x^2 - \hat{S}_y^2), \quad (13)$$

With the $\hat{S}_x, \hat{S}_y$ Cartesian spin operators being related to the raising and lowering operators via

$$\hat{S}_+ |s, m\rangle = \frac{1}{2} (\hat{S}_x + i\hat{S}_y) |s, m\rangle \quad (14)$$

$$= \sqrt{(s(s+1) - m(m+1))} |s, m+1\rangle. \quad (15)$$

$$\hat{S}_- |s, m\rangle = \frac{1}{2} (\hat{S}_x - i\hat{S}_y) |s, m\rangle \quad (16)$$

$$= \sqrt{(s(s+1) - m(m-1))} |s, m-1\rangle. \quad (17)$$

By rearranging Equation 14 in terms of $\hat{S}_x, \hat{S}_y$, we get

$$\hat{S}_x = \frac{1}{2} (\hat{S}_+ + \hat{S}_-) \quad (18)$$

$$\hat{S}_y = \frac{1}{2i} (\hat{S}_+ - \hat{S}_-) \quad (19)$$

$$\hat{S}_x^2 + \hat{S}_y^2 = \hat{S}_x^2 - \hat{S}_y^2 = \frac{1}{2} (\hat{S}_+^2 + \hat{S}_-^2). \quad (20)$$

Therefore, a general expression for the Zero-Field Splitting Hamiltonian acting on $|s, m\rangle$ can be written as

$$\begin{aligned}\hat{H}_{ZFS}|s, m\rangle &= Dm^2|s, m\rangle - \frac{D}{3}(s(s+1))|s, m\rangle \\ &+ \frac{E}{2}(s(s+1) - m(m+1))|s, m+2\rangle \\ &+ \frac{E}{2}(s(s+1) - m(m-1))|s, m-2\rangle.\end{aligned}$$

4 Exchange Interaction

Consider two electrons in orbitals $\phi_a(\vec{r}), \phi_b(\vec{r})$. The Hamiltonian is

$$\hat{H} = \sum_{i=1}^2 \left[-\frac{\hbar^2}{2m} \nabla_i^2 + V(\vec{r}_i) \right] + \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}.$$

Here, $V(\vec{r})$ is the single-particle potential. Because electrons are fermions, their total state must be antisymmetric under spatial and spin exchange. We factorize into spatial $\Psi(\vec{r}_1, \vec{r}_2)$ and spin $\chi(s_1, s_2)$ parts:

$$\Psi_{\text{tot}}(1, 2) = \Psi(\vec{r}_1, \vec{r}_2)\chi(s_1, s_2),$$

with

$$\Psi_{\text{tot}}(2, 1) = -\Psi_{\text{tot}}(1, 2).$$

If Ψ is symmetric, then χ must be antisymmetric (singlet). If Ψ is antisymmetric, χ must be symmetric (triplet).

We build orthonormal spatial orbitals ϕ_a, ϕ_b . Two natural two-electron spatial states are:

$$\begin{aligned}\Psi_S(\vec{r}_1, \vec{r}_2) &= \frac{1}{\sqrt{2}} [\phi_a(1)\phi_b(2) + \phi_b(1)\phi_a(2)] \\ \Psi_A(\vec{r}_1, \vec{r}_2) &= \frac{1}{\sqrt{2}} [\phi_a(1)\phi_b(2) - \phi_b(1)\phi_a(2)].\end{aligned}$$

The corresponding spin states are

$$\begin{aligned}\chi_S &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (\text{singlet}) \\ \chi_A &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad (\text{triplet}).\end{aligned}$$

We diagonalize $\hat{H}$ into the two-dimensional subspace spanned by $\phi_a\phi_b$. Because $\hat{H}_0$ acts on each electron separately and both spatial states Ψ_S, Ψ_A have the same one-particle energy,

$$E_0 = \langle \phi_a | \hat{h} | \phi_a \rangle + \langle \phi_b | \hat{h} | \phi_b \rangle, \quad \hat{h} = -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r});$$

their splitting arises entirely from the coulomb term:

$$\hat{H}_C = \frac{e^2}{4\pi\epsilon_0 r_{12}}$$

Defining the direct and exchange integrals:

$$K \equiv \int \int d^3r_1 d^3r_2 |\phi_a(1)|^2 \frac{e^2}{4\pi\epsilon_0 r_{12}} |\phi_b(2)|^2,$$

$$J \equiv \int \int d^3r_1 d^3r_2 \phi_a^*(1) \phi_b(1) \frac{e^2}{4\pi\epsilon_0 r_{12}} \phi_b^*(2) \phi_a(2).$$

We find

$$\langle \Psi_S | \hat{H} | \Psi_S \rangle = E_0 + K + J,$$

$$\langle \Psi_A | \hat{H} | \Psi_A \rangle = E_0 + K - J.$$

Thus the singlet sits at energy $E_S = E_0 + K + J$, while each triplet has $E_T = E_0 + K - J$. Their energy difference/splitting is

$$\Delta E \equiv E_S - E_T = 2J.$$

The energy splitting due to $\hat{H}_{ZFS}$ is analogously also $2D$. Regardless, we now wish to replace this two level spatial+spin system by a purely spin Hamiltonian that reproduces the same energy shift between singlet and triplet.

Recall that the operator exchanging two spins is

$$\hat{P}_{12}\chi(1,2) = \chi(2,1),$$

It's eigenvalues are +1 on the symmetric (triplet) subspace and -1 on the antisymmetric (singlet) subspace. Define the total spin operator

$$\hat{S}_{\text{tot}} = \hat{S}_1 + \hat{S}_2,$$

and its square

$$\hat{S}_{\text{tot}}^2 = (\hat{S}_1^2 + \hat{S}_2^2) + \hat{S}_1^2 + \hat{S}_2^2 + 2\hat{S}_1 \cdot \hat{S}_2.$$

For spin- $\frac{1}{2}$, $\hat{S}_1^2 = \hat{S}_2^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = \frac{3\hbar^2}{4} \mathbf{1}$. Hence,

$$\hat{S}_{\text{tot}}^2 = \frac{3\hbar^2}{2} \mathbb{1} + 2\hat{S}_1 \cdot \hat{S}_2. \quad (21)$$

Now, for both triplet ($S_{\text{tot}} = 1$) and singlet ($S_{\text{tot}} = 0$) states,

$$\begin{aligned} \hat{S}_{\text{tot}}^2|10\rangle &= S(S+1)\hbar^2|10\rangle = 2\hbar^2|10\rangle & \text{We want } \hat{P}_{12}|10\rangle &= +|10\rangle \\ \hat{S}_{\text{tot}}^2|00\rangle &= S(S+1)\hbar^2|00\rangle = 0 & \text{We want } \hat{P}_{12}|00\rangle &= -|00\rangle \end{aligned}$$

We define an ansatz $\hat{P}_{12} = a\hat{S}_{\text{tot}}^2 + b\mathbb{1}$. Hence

$$\begin{aligned} \hat{P}_{12}|10\rangle &= a(2) + b|10\rangle = +1|10\rangle \\ \hat{P}_{12}|00\rangle &= a(0) + b|00\rangle = -1|00\rangle. \end{aligned}$$

Solving,

$$b = -1, \quad 2a - 1 = +1 \Rightarrow 2a = 2 \Rightarrow a = 1.$$

Therefore,

$$\hat{P}_{12} = 1 \cdot \hat{S}_{\text{tot}}^2 - 1 \cdot \mathbb{1} = \hat{S}_{\text{tot}}^2 - \mathbb{1}.$$

Substituting back in $\hat{S}_{\text{tot}}^2$ from Equation 21,

$$\hat{P}_{12} = \left(\frac{3}{2} \mathbb{1} + 2\hat{S}_1 + \hat{S}_2 \right) - \mathbb{1} = \frac{1}{2} \mathbb{1} + 2\hat{S}_1 \cdot \hat{S}_2.$$

We get the final expression for the exchange operator:

$$\hat{P}_{12} = \frac{1}{2} \left(1 + 4\hat{S}_1 \cdot \hat{S}_2 \right).$$

It's eigenvalues are -1 on the singlet and +1 on the triplet manifold. Therefore an operator proportional to $\hat{P}_{12}$ will split singlet/triplet. Define the proportionality coefficient J_{ex} such that

$$\hat{H}_{EX} = J_{ex} \hat{P}_{12} = \frac{J_{ex}}{2} \left(1 + 4\hat{S}_1 \cdot \hat{S}_2 \right).$$

Therefore, acting on

$$\begin{aligned} \chi_S : \hat{H}_{EX} \chi_S &= J_{ex}(-1) \chi_S \\ \chi_T : \hat{H}_{EX} \chi_T &= J_{ex}(+1) \chi_T. \end{aligned}$$

To reproduce $\Delta E = E_S - E_T = 2J$, we set $J_{ex} = -J$. Then,

$$E_S = -J_{ex} = +J, \quad E_T = +J_{ex} = -J,$$

as desired. Setting J to be the exchange coupling matrix proportional to the degree of overlap between wavefunctions, the exchange Hamiltonian can be expressed as

$$\hat{H}_{EX} = \hat{\vec{S}}_a \cdot J \cdot \hat{\vec{S}}_b.$$

If we assume the exchange coupling is isotropic, we can simplify to

$$\hat{H}_{EX} = -J \hat{\vec{S}}_a \cdot \hat{\vec{S}}_b.$$

5 Putting it all together

For a two-electron (carrier + defect) and two-nuclei (Silicon and Carbon) spin system, there are 16 orthonormal basis states that make up the 16×16 Spin Hamiltonian. Written in the coupled-electron $|s, m\rangle$ basis tensored with each nucleus m_I eigenstate:

Triplet $|1, +1\rangle$ manifold

$$\begin{aligned} &|1, 1\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, 1\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \\ &|1, 1\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, 1\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \end{aligned}$$

Triplet $|1, 0\rangle$ manifold

$$\begin{aligned} &|1, 0\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, 0\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \\ &|1, 0\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, 0\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \end{aligned}$$

Singlet $|0, 0\rangle$ manifold

$$\begin{aligned} &|0, 0\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|0, 0\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \\ &|0, 0\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|0, 0\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \end{aligned}$$

Triplet $|1, -1\rangle$ manifold

$$\begin{aligned} &|1, -1\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, -1\rangle \otimes |+\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \\ &|1, -1\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |+\frac{1}{2}\rangle_2 \\ &|1, -1\rangle \otimes |-\frac{1}{2}\rangle_1 \otimes |-\frac{1}{2}\rangle_2 \end{aligned}$$

Our goal is to now programmatically calculate the Spin Hamiltonian's effect on each coupled-basis state above. To do this, we calculate the Zeeman, Hyperfine, ZFS, and Exchange Hamiltonian separately in the coupled basis, and then sum them together.